

IFEval Factor Interpretation Under a Product-Mixture Probit Factor Model

Threshold 2/3; full-data refit

1 Fitted Model

This note summarizes the full-data IFEval fit at threshold 2/3, where an item is scored as correct when the OpenEval strict-instruction score is at least 2/3. The fitted model is the sparse product-mixture probit factor model with

$$H = 5, \quad G = (3, 3, 2, 3, 3), \quad \lambda_{\ell_1} = 4.$$

The binary response matrix contains 122 models and 538 nonconstant IFEval items. The converged full-data refit used sampled probit augmentation, 14 pretraining iterations, and 11 MAP refinement iterations. All marginal mixtures converged in the selected refinement iteration. The fitted binary log likelihood per observed response was -0.246 .

Each factor coordinate is standardized by the fitted product-mixture scale convention. Larger positive scores therefore mean stronger performance on items with positive loadings on that factor, after accounting for the other coordinates and item intercepts.

2 Loading Structure

Figure 1 shows the sparse loading matrix after sorting items by their largest absolute loading. The first factor is broad: most items have their largest loading on F1. The remaining factors carve out more local modes of instruction-following difficulty. The item counts by primary loading are: F1 = 391, F2 = 65, F3 = 35, F4 = 34, and F5 = 13.

Largest refined item loadings

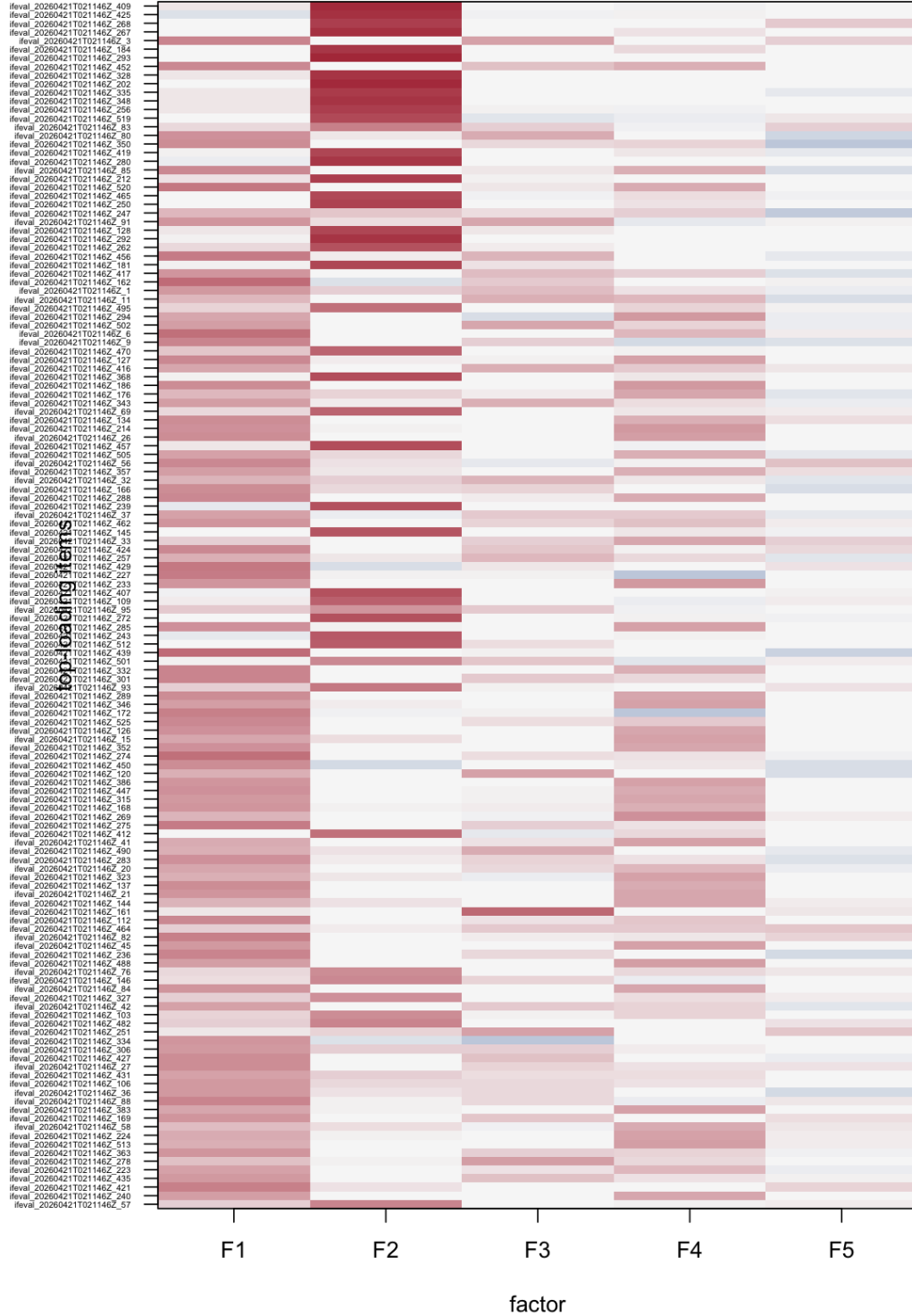


Figure 1: Estimated sparse loading matrix for the selected H=5 IFEval fit. Rows are items, columns are factors, and rows are grouped by the factor with largest absolute loading.

3 Factor Interpretation

The factors are best read as overlapping psychometric modes rather than as the 25 raw IFEval instruction categories. The model compresses those instruction checks into five lower-dimensional axes:

Factor	Label	Interpretation
F1	Compositional response control	Broad ability to satisfy multi-part prompts, repeated-prompt requirements, sectioning, highlighting, paragraph structure, and several constraints at once.
F2	Case and language mode control	Ability to stay in a required surface mode, especially all-lowercase, all-uppercase, and specified response language.
F3	Lexical micro-control	Fine token-level and count-based control: forbidden letters or words, letter frequency, sentence counts, and unusually precise lexical inhibition.
F4	Boundary and envelope formatting	Formatting wrappers and response boundaries, including JSON, quotation marks, and explicit ending constraints.
F5	Length and count discipline	Short-response length control, word limits, sentence limits, and count-based keyword requirements.

The instruction enrichment plots in Figure 2 support this interpretation. F2 is highly enriched for case and language instructions; F3 for letter-frequency and exact lexical/count constraints; F4 for JSON and start/end wrappers; and F5 for compact length-control items.

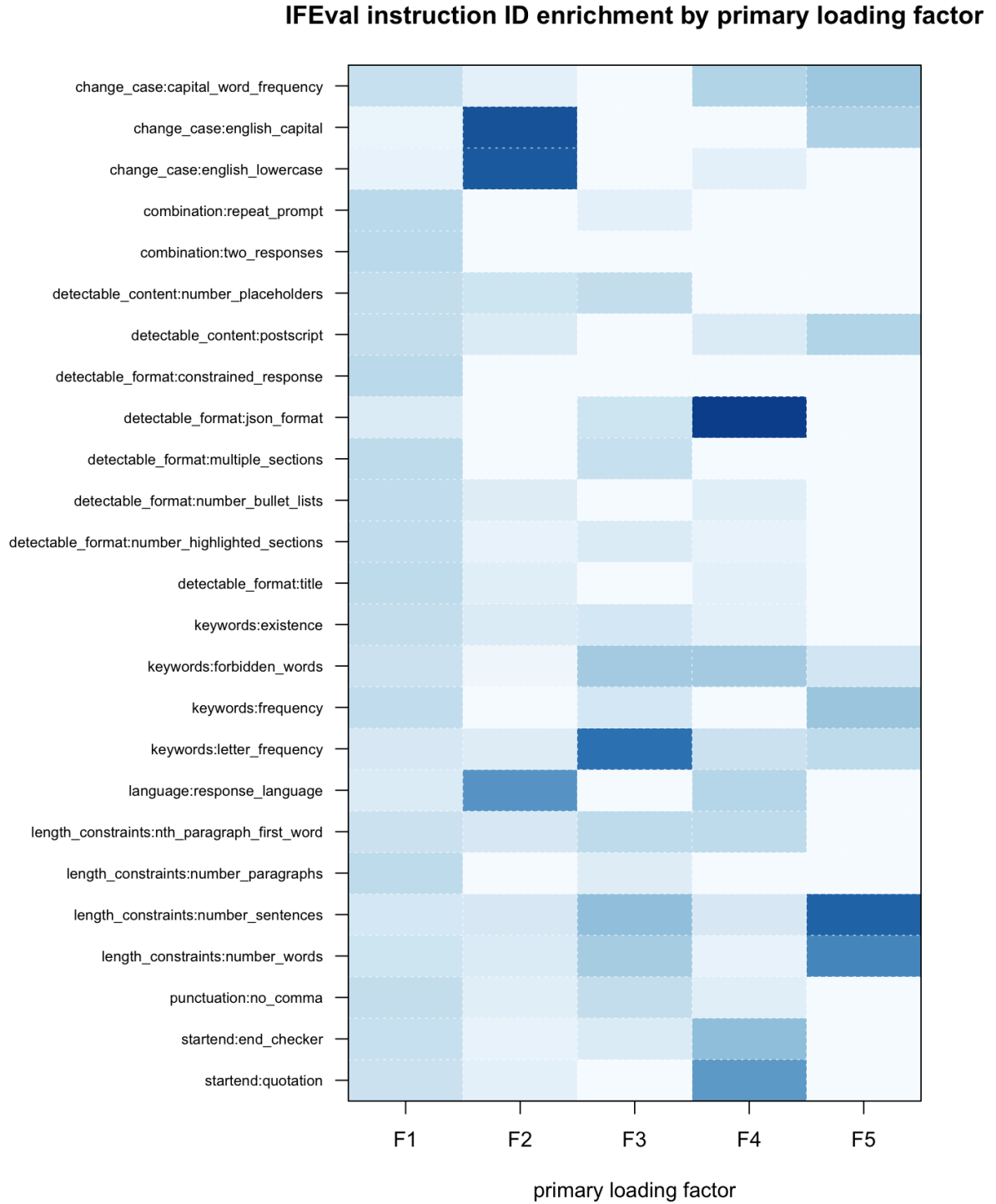


Figure 2: Instruction-ID enrichment by primary factor. Darker cells indicate larger lift relative to the overall frequency of that instruction.

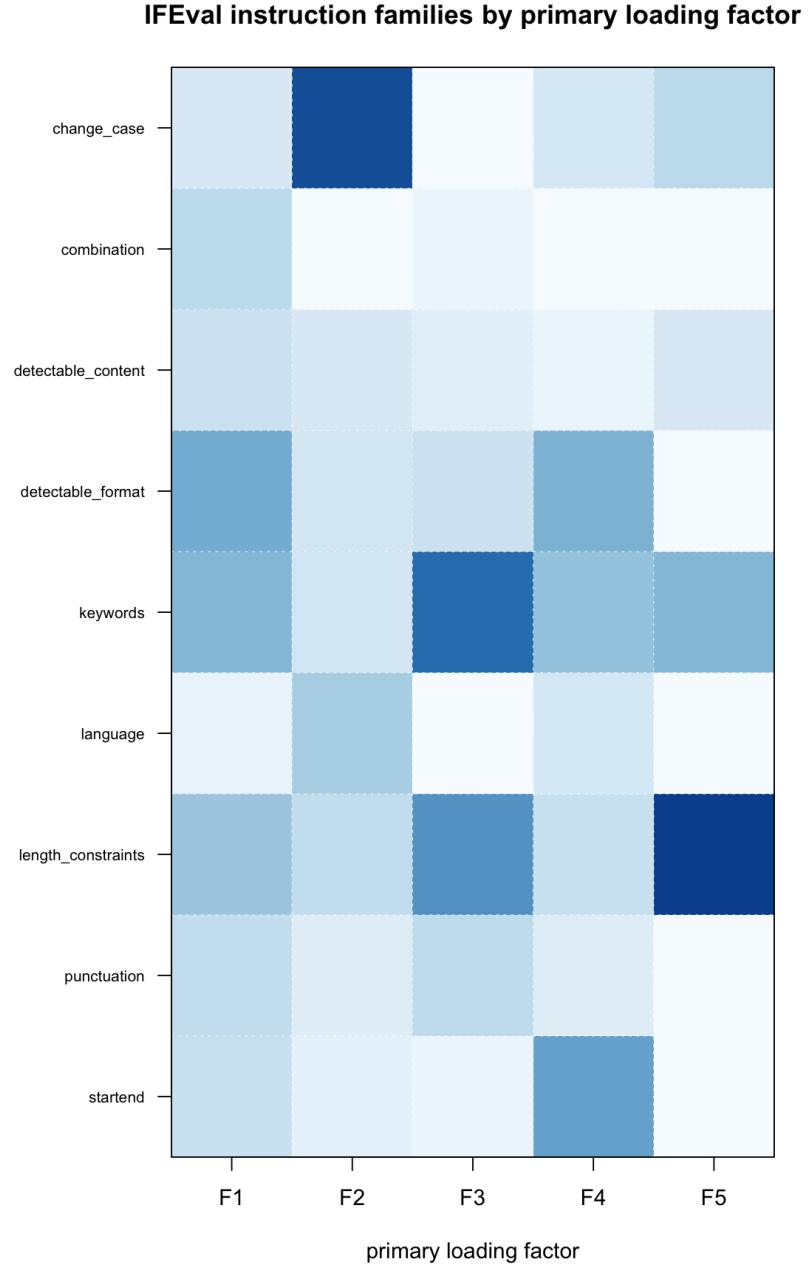


Figure 3: Instruction-family proportions within each primary factor group.

4 Strong Solo-Loading Items

Table 1 gives examples whose primary loading is large and whose second-largest absolute loading is comparatively small. These are clean representatives of each factor, although the factor labels should also be read with the enrichment plots above.

Table 1: Strong mostly solo-loading item examples. Accuracy is empirical accuracy at the 2/3 threshold.

Factor	Item	Loading	Prompt excerpt
F1	...274	1.779	Give two different responses to “Is it ethical to hunt and eat invasive species?”, separated by six asterisks and without using any commas.
F1	...242	1.759	First repeat the request word for word, then give the answer; include a title wrapped in double angular brackets.
F2	...293	3.032	Create a product description for a product that helps stop snoring. Use all lowercase letters.
F2	...409	2.969	Explain the difference between two-stroke and four-stroke motors; the entire response must be in English and contain only lowercase letters.
F3	...161	1.895	Continue and expand a scene description using at least 60 sentences but fewer than 600 words.
F3	...277	1.677	Write a rubric as a poem, with the letter “w” appearing fewer than two times.
F4	...50	0.562	Write a haiku about rushing to work using only Marathi, with no other language allowed.
F4	...217	0.400	Rewrite a research-paper summary in APA format while avoiding specified words.
F5	...75	0.522	Write a funny song with fewer than 10 sentences for a playground proposal.
F5	...232	0.470	Write a template with fewer than seven sentences for calculating an array offset.

5 Cross-Loading Items

Cross-loading items are especially useful because they show why the factors are not simply a relabeling of the raw instruction IDs. A prompt can require a general compositional skill plus one local control skill, or two local skills at once.

Table 2: Representative cross-loading item examples.

Pair	Primary loading	Secondary loading	Prompt excerpt
F1/F4	F1 = 1.317	F4 = 1.230	Give an opinion on Kareena Kapoor and wrap the entire response in double quotation marks.
F1/F3	F1 = 1.497	F3 = 1.091	Write a resignation email, repeat the request verbatim first, and obey additional lexical restrictions.
F1/F5	F1 = 1.132	F5 = -0.785	Give advice about a dime in the style of a U.S. president and make the response at least 600 words.

Pair	Primary loading	Secondary loading	Prompt excerpt
F3/F4	F3 = 0.873	F4 = 0.765	Create a product advertisement, with the entire output in JSON format.
F2/F5	F2 = 1.176	F5 = 0.686	Explain the difference between a levee and an embankment, responding only in Korean.
F1/F2	F1 = 0.888	F2 = 0.663	Write a long funny sales pitch while satisfying a word-length target and other surface-form constraints.
F3/F5	F3 = 1.213	F5 = -0.618	Write an investor-facing essay about the history of the Internet with at least 50 sentences.
F2/F3	F2 = 1.418	F3 = 0.580	Write a professional but weird cover letter with a strict letter-frequency constraint.
F2/F4	F2 = 1.772	F4 = 0.376	Write an all-capital, coach-like article with no lowercase letters.

6 Marginal Mixtures and LLM Profiles

Figure 4 shows the fitted one-dimensional mixtures for the five factor scores. The fitted groups are ordered by their means within each coordinate, so cluster labels are coordinate-specific. For example, a high cluster on F1 is not the same population subset as a high cluster on F3.

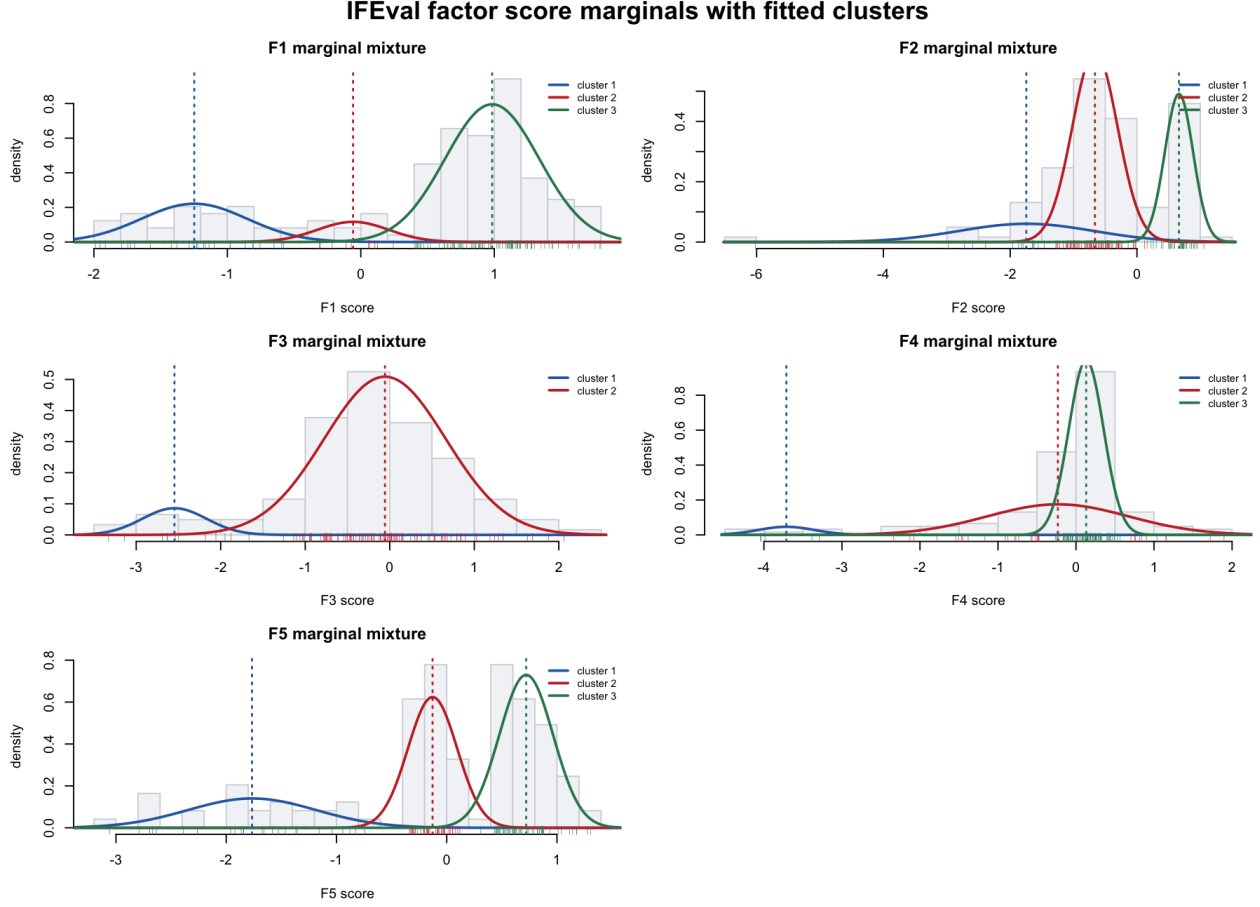


Figure 4: Estimated marginal distributions of the five factor coordinates with fitted mixture components overlaid.

The joint profiles give a compact description of model behavior. The strongest overall models, such as `gpt-5.4-pro-2026-03-05`, `gpt-5.2-pro-2025-12-11`, and `grok-4-0709`, tend to be high on F1 and not weak on the local factors. Some models have uneven profiles: `gpt-5-2025-08-07` is very high on F3 but low on F2, suggesting strong lexical/count control but weaker case/language mode control in this fitted coordinate system. Smaller or older models often have low F1 and weak local coordinates; this predicts failures on prompts that combine several constraints with exact formatting, casing, or count requirements.

This is the main psychometric value of the model: it does not merely rank models by average IFEval accuracy. It separates broad compositional control from local failure modes, and it gives each LLM a joint ability profile that can be linked back to concrete prompt types.

7 Takeaway

The $H=5$, $G = (3, 3, 2, 3, 3)$ fit suggests that IFEval is much lower-dimensional than its 25 instruction IDs, but not one-dimensional. F1 captures broad multi-constraint response control, while F2–F5 isolate specific surface-form, lexical, boundary-formatting, and length/count abilities. Cross-loading items are not nuisances: they are the items that reveal how these skills interact in real instruction-following prompts.